Assignment No. 2

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Q1. Given sentence: "The bank will not open until the river recedes."

a) Identify and explain the ambiguity

Ans: The ambiguous word in the sentence is "bank."

- **Bank (financial institution):** A financial place where people deposit, keep, or withdraw money.
- **Bank (river bank):** The sloping land directly beside a body of water or river.

Here, the correct meaning intended by context is river bank, because the sentence explicitly refers to the river receding.
Type: Lexical ambiguity / Word Sense Ambiguity.

b) How WordNet or BabelNet can help

Ans: WordNet lists different distinct meanings (known as synsets) of the word bank and provides semantic relationships (hypernyms, hyponyms) between words and their meanings.
BabelNet combines WordNet-like taxonomic information with multilingual encyclopedic knowledge from sources like Wikipedia.
The automated system can compare surrounding contextual words such as "river" and "recedes" with the glosses and semantic relations of the possible senses of bank. By calculating contextual overlap, it selects bank = land beside a river.

c) Why automatic systems may still fail

Ans: Automatic NLP systems can still fail due to the following factors:

- Context may be too short, sparse, or ambiguous to provide enough evidence.
- A word can have several fine-grained or closely related senses that are hard to distinguish.
- Dictionaries and knowledge bases may lack sufficient contextual or domain-specific coverage.
- Complex interpretation often requires real-world common-sense knowledge.
- Different specialized domains use identical words in highly specific, unconventional ways.

Q2. Naïve Bayes for Word Sense Disambiguation

a) How Naïve Bayes is applied to WSD

Ans: Naïve Bayes classifies a target ambiguous word into one of its possible pre-defined senses by evaluating surrounding context words. For example, for "bank":

- Sense 1 (S1): Financial institution
- Sense 2 (S2): River side

The Bayes classifier calculates the posterior probability for each sense given the context features:

$$P(Sense|Context) \propto P(Sense) \times \prod P(Word_i|Sense)$$

The sense yielding the highest calculated probability is assigned. For instance, in "The fisherman sat on the bank of the river;" words like fisherman, river, and sat strongly maximize the probability for the river-bank sense.

b) Limitations compared to Lesk's approach

Feature/Dimension	Naïve Bayes Approach	Lesk's Approach
Training Data	Requires supervised sense-annotated training data.	Mainly knowledge/dictionary based (unsupervised).
Learning Method	Learns probabilistic weights from training examples.	Uses dictionary definitions and gloss overlaps.
Core Mechanism	Assumes context features are conditionally independent.	Compares semantic overlap between glosses and context.
Performance Basis	Performance depends heavily on quality of training data.	Can function effectively with limited training data.
Unseen Contexts	May perform poorly on out-of-domain/unseen context.	Leverages dictionary definitions dynamically.
Domain Adaptability	Requires domain-specific re-training.	Uses general dictionary overlap definitions.

c) Improvement using Yarowsky's algorithm

Ans: Yarowsky's algorithm is a bootstrapping (semi-supervised) approach:

- Start with a small set of seed rules / labelled examples.
- Train an initial classifier and predict senses on unlabelled data.
- Add the highest-confidence predictions to the training set and retrain iteratively.
- **Benefit:** Significantly reduces the need for large hand-annotated datasets while improving coverage.

Q3. Given sentence: "John met Alex after he finished the project."

a) Identify the anaphora and explain the difficulty

Ans: The anaphoric expression is the pronoun "he."
Candidate antecedents: **John** and **Alex** (both are singular male entities).
Difficulty: Both candidates satisfy gender and number agreement rules, so syntax alone cannot determine who finished the project.

b) How would Hobbs' algorithm resolve "he"?

Ans: Hobbs' algorithm traverses the syntactic parse tree using a deterministic order:

1. Start at the pronoun node ("he") and ascend to the nearest NP/S ancestor.
2. Traverse left-to-right to collect candidate Noun Phrases matching gender (Male) and number (Singular).
3. Standard parse tree traversal order generally favors the main subject (**John**).

c) Real-world scenario where syntax alone fails

Ans: "The trophy did not fit in the suitcase because it was too large."
Syntactically, "it" could refer to trophy or suitcase, but real-world knowledge is required to deduce that the trophy is too large.

Conclusion

> Word Sense Disambiguation (WSD) and Anaphora Resolution demonstrate core natural language processing challenges. While computational models like Naïve Bayes classifiers and Hobbs' parse-tree algorithm offer structured ways to address ambiguity, fully accurate resolution requires combining dictionary resources (WordNet/BabelNet) with real-world contextual semantics.